

Module 7

Financial Management

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

$$\begin{aligned} E(R_A) &= 0.2 \times -8\% + 0.5 \times 10\% + 0.3 \times 15\% \\ &= 7.9\% \end{aligned}$$

$$\begin{aligned} E(R_B) &= 0.2 \times 30\% + 0.5 \times 15\% + 0.3 \times -12\% \\ &= 9.9\% \end{aligned}$$

Answer 1(b)

$$\begin{aligned} \sigma_A^2 &= 0.2 \times (-0.08 - 0.079)^2 + 0.5 \times (0.10 - 0.079)^2 + 0.3 \times (0.15 - 0.079)^2 \\ &= 0.006789 \\ \sigma_A &= 8.24\% \end{aligned}$$

$$\begin{aligned} \sigma_B^2 &= 0.2 \times (0.3 - 0.099)^2 + 0.5 \times (0.15 - 0.099)^2 + 0.3 \times (-0.12 - 0.099)^2 \\ &= 0.023769 \\ \sigma_B &= 15.42\% \end{aligned}$$

Answer 1(c)

Let W_A and W_B be the weights of stock A and stock B, respectively.

$$E(R_P) = W_A \times E(R_A) + (1 - W_A) \times E(R_B)$$

$$\begin{aligned} W_A &= (E(R_P) - E(R_B)) / (E(R_A) - E(R_B)) \\ &= (0.09 - 0.099) / (0.079 - 0.099) \\ &= -0.009 / -0.02 \end{aligned}$$

$$W_A = 0.45$$

$$W_B = 1 - 0.45 = 0.55$$

Answer 1(d)

Correlation coefficient between stock A and stock B = $\rho = \text{COV}_{R_{AB}} / (\sigma_A \times \sigma_B)$

$$\begin{aligned} \text{COV}_{R_{AB}} &= 0.2 \times (-0.08 - 0.079) \times (0.3 - 0.099) + 0.5 \times (0.10 - 0.079) \times (0.15 - 0.099) + 0.3 \\ &\times (0.15 - 0.079) \times (-0.12 - 0.099) \\ &= -0.010521 \end{aligned}$$

$$\begin{aligned} \rho &= -0.010521 / (0.0824 \times 0.1542) \\ &= -0.83 \end{aligned}$$

Answer 1(e)(i)

The statistical measure that identifies how asset returns move in relation to each other is known as the correlation coefficient:

The value ranges of the correlation coefficient is between -1 and 1.

A negative correlation indicates that the returns are moving in opposite directions.

If its value is -1, it means that the two stocks are perfectly negatively correlated (i.e., as one goes up, the other goes down);

A positive correlation indicates movement in the same direction. If its value is +1, it means that the two stocks are perfectly positively correlated (i.e., both go up/ down together.);

If the value is 0, it means that the two stocks are unrelated.

Answers 1(e)(ii)

If the correlation between two stocks is imperfect, meaning that the correlation coefficient is less than 1, there is a benefit from diversifying by holding two stocks with different risk characteristics.

In a two-stock portfolio, if $\rho = -1$, the two stocks should be combined so that there will be maximum diversification of the portfolio.

On the other hand, if $\rho = +1$, there is no diversification potential when combining the two stocks.

In general, if ρ is between +1 and -1, it means the risk of the portfolio is lowered but not eliminated.

Answer 2(a)

$$R_E = \text{DIV}_E / P_0 + g$$

$$\text{DIV}_E = \text{HK\$1}$$

$$g = 0.03$$

$$P_0 = \text{HK\$10}$$

$$R_E = \text{HK\$1} / \text{HK\$10} + 0.03$$

$$= 0.1 + 0.03$$

$$= 0.13$$

Answer 2(b)

$$\text{Cost of debt} = R_D = \text{YTM} = 6.5\%$$

$$R_D \text{ after tax} = 6.5\% \times (1 - 16.5\%) = 5.43\%$$

$$\text{Cost of preferred stock} = R_{ps} = \text{Div}_{ps} / P_{ps}$$

$$\text{Div}_{ps} = 6\% \times 100 = \text{HK\$6}$$

$$P_{ps} = \text{HK\$86}$$

$$R_{ps} = \text{HK\$6} / \text{HK\$86} = 6.98\%$$

Answer 2(c)

Market value of common stock: $E = 50 \text{ million} \times \text{HK\$}10 = \text{HK\$}500 \text{ million}$

Market value of debt: $D = 300,000 \times \text{HK\$}820 = \text{HK\$}246 \text{ million}$

Market value of preferred stock: $P = 3 \text{ million} \times \text{HK\$}86 = \text{HK\$}258 \text{ million}$

Total market value:

$V = \text{HK\$}500 \text{ million} + \text{HK\$}246 \text{ million} + \text{HK\$}258 \text{ million} = \text{HK\$}1,004 \text{ million}$

$W_E = E / V = \text{HK\$}500 \text{ million} / \text{HK\$}1,004 \text{ million} = 49.80\%$

$W_P = P / V = \text{HK\$}246 \text{ million} / \text{HK\$}1,004 \text{ million} = 24.50\%$

$W_D = D / V = \text{HK\$}258 \text{ million} / \text{HK\$}1,004 \text{ million} = 25.70\%$

$\text{WACC} = 49.80\% \times 13.0\% + 24.50\% \times 5.43\% + 25.70\% \times 6.98\% = 9.60\%$

Answer 2(d)

NPV(A)

$= - \text{HK\$}1,200,000 + \text{HK\$}400,000 / 1.096 + \text{HK\$}450,000 / 1.096^2 + \text{HK\$}420,000 / 1.096^3$
 $+ \text{HK\$}420,000 / 1.096^4 + \text{HK\$}200,000 / 1.096^4 = \text{HK\$}288,288$

NPV(B)

$= - \text{HK\$}1,100,000 + \text{HK\$}200,000 / 1.096 + \text{HK\$}350,000 / 1.096^2 + \text{HK\$}450,000 / 1.096^3$
 $+ \text{HK\$}450,000 / 1.096^4 + \text{HK\$}400,000 / 1.096^4$
 $= \text{HK\$}304,743$

Venus should accept order B because it has a higher NPV.

Answer 2(e)

Payback period for order A

$= 2 + \text{HK\$}350,000 / \text{HK\$}420,000 = 2.83$

Payback period for order B

$= 3 + \text{HK\$}100,000 / \text{HK\$}450,000 = 3.22$

Venus should accept order A because it has a shorter payback period.

Disadvantages:

- It does not consider the cash flows after the breakeven point.
- It does not consider the timing of cash flows within the payback period.
- The choice of the required payback period by the company is arbitrary.
- It does not quantify the changes in shareholder values.

Answer 3(a)

Key performance indicators (KPIs) are values that can be identified, measured, and used to track whether a company's implemented strategy achieved its mission successfully. Choosing KPIs to monitor processes is necessary for the successful implementation of a business strategy.

The following elements are crucial in the development of a set of good KPIs.

- Easy to understand: The metrics should be as simple as possible.
- Readily quantifiable: An indicator should be able to be measured easily; otherwise it may not be useful.
- Easy to monitor: If KPIs are difficult or expensive to monitor, they may not be useful.
- Linked to company strategy: Indicators must be related to the company's strategy and/ or objectives; otherwise they may not serve a clear purpose.

Answer 3(b)

The four perspectives of a balanced scorecard are as follows:

- Financial perspective: Financial measures show the impact on the bottom line of the strategy implemented by the company.
- Customer perspective: It includes and reflects customer perceptions with measures which show how companies are providing value to their customers.
- Internal process perspective: For an effective company, internal business processes should be managed effectively and efficiently.
- Learning and growth perspective: Organisational learning and innovation are important to remain competitive and tie directly to the company's value.

Answer 3(c)

The suggested KPIs and their values for Pluto are as follows:

- Financial perspective:
 $\text{Gross Profit (existing products)} = (\text{HK\$}265,500 - \text{HK\$}215,055) / \text{HK\$}265,500 = 19.0\%$
- Customer perspective:
 $\text{Percentage of new customers} = 15 / (203 + 15) = 6.88\%$
- Internal process perspective:
 $\text{Productivity (units per direct labour hour) – existing product} = 13,000 / 1,936 = 6.71 \text{ unit/ hr}$
- Learning and growth perspective:
 $\text{Development costs as \% of sales} = \text{HK\$}15,075 / (\text{HK\$}265,500 + \text{HK\$}37,500) = 4.98\%$

Answer 3(d)

The advantages of using a balanced scorecard for performance measurement are as follows:

- They are closely linked to short-term and long-term strategies of the company.
- Success drivers are often the intangibles that are not reflected in the financial statements but may be tracked by non-financial measures of a balanced scorecard.
- Non-financial measures may be critical to the future performance of the company.
- Managers can choose non-financial measures which are within their controls.

The disadvantages of using a balanced scorecard for performance measurement are as follows:

- Tracking the non-financial measures of a balanced scorecard maybe costly and time-consuming and is often greater than the benefits doing so would generate.
- There is also a danger that measuring can become mechanical and, consequently, may not be useful.
- Non-financial measures may be subjective, which may make it very difficult to determine whether the objectives of the company can be achieved or not.
- Non-financial measures are not statistically reliable.

Answer 4(a)

Profitability

2022

Gross profit margin = $\text{HK\$}470 / \text{HK\$}2,050 = 22.93\%$

Return on equity = $\text{HK\$}165 / \text{HK\$}1,460 = 11.30\%$

2023

Gross profit margin = $\text{HK\$}520 / \text{HK\$}2,230 = 23.32\%$

Return on equity = $\text{HK\$}149 / \text{HK\$}1,564 = 9.53\%$

Liquidity

2022

Current ratio = $\text{HK\$}609 / \text{HK\$}525 = 1.16$

Quick ratio = $(\text{HK\$}609 - \text{HK\$}375) / \text{HK\$}525 = 0.45$

2023

Current ratio = $\text{HK\$}822 / \text{HK\$}544 = 1.51$

Quick ratio = $(\text{HK\$}822 - \text{HK\$}486) / \text{HK\$}544 = 0.62$

Gearing

2022

Debt to equity ratio = $\text{HK\$}(800 + 525) / 1,460 = 90.8\%$

Interest coverage ratio = $\text{HK\$}243 / \text{HK\$}45 = 5.40$

2023

Debt to equity ratio = $\text{HK\$}(1,000 + 544) / 1,564 = 98.7\%$

Interest coverage ratio = $\text{HK\$}235 / \text{HK\$}56 = 4.20$

Answer 4(b)

Profitability

Net sales increased by 8.8% and the gross profit margin has also had a marginal increase of 0.4% (from 22.9% to 23.3%). It seems that Jupiter still can maintain a gross profit margin while achieving a reasonable growth rate in sales. However, there was a drop in net profit (from HK\$165 million to HK\$149 million) which may be a concern for Jupiter. Furthermore, the return on equity shows a drop from 11.3% to 9.5%. It should be noted that the drop in both net profit and ROE are due to the large increases in operating costs, depreciation, and interest expenses. Since the increases in depreciation and interest expenses could not be avoided due to the large capital investment in 2023 to upgrade the production capacity, Jupiter should investigate the reasons for the significant increase in operation expenses (such as staff remuneration and distribution costs) to see whether there is room for improvement.

Liquidity

The current ratio for Jupiter increased from 1.16 to 1.51, but this may still not be regarded as a healthy level as compared to the general target current ratio of 2.0. However, the acceptability of the ratio will depend on the industry that Jupiter operates in and how it compares to those of peer companies. In addition, the quick ratio does not look good for Jupiter, although there is a slightly improvement from 0.45 to 0.62. The ratio is much lower than the general target quick ratio of 1.0. When we look at the quick ratio for Jupiter, we might be concerned about the liquidity of the company. As Jupiter carries a significant amount of inventory, the quick ratio is very low, at 0.62 times (even if this is up from 0.45). It is suggested that the liquidity position inventory management should be improved to reduce the inventory level.

Gearing

In terms of gearing, the interest coverage ratio has gone down from 5.4 to 4.2. The major reasons for the decrease in interest coverage ratio are: (a) an increase in interest expense due to an additional bank loan for capital investment; and (b) an increase in operating expenses which reduced the profit before interest and tax. In addition, the debt-to-equity ratio increased from 90.8% to 98.7%. The increase in this ratio is mainly due to the additional bank loan which increased the overall debt level of the company. However, Jupiter appears to have its capital structure under good control. The current gearing ratios are still maintained at an acceptable level. It should be noted that the deterioration in gearing ratios is mainly due to the need to increase the production capacity, which should have positive effects on the long-term financial performance of the company. As such, it is anticipated that the gearing ratios will be able to be improved in the future.

Answer 4(c)

Besides return on investment (ROI), two common financial measures for responsibility centres are:

Residual Income (RI)

Residual income (RI) measures the amount of profits exceeding a required rate of return, which is also known as capital charge.

The RI is calculated as:

$$\text{RI} = \text{Operating income} - (\text{Required rate of return} \times \text{Average operating assets})$$

Many companies have minimum expectations of the values of RI for their operations and investments.

Economic Value added (EVA)

Economic value added (EVA) is a type of residual income incorporating several adjustments to lower some of the disadvantages created by residual income.

The EVA is calculated as:

$$\text{EVA} = \text{Adjusted after-tax operating income} - [\text{WACC} \times (\text{Adjusted total assets} - \text{Current liabilities})]$$

EVA solved the problems that financial measures based on profits calculated in accordance with accounting standards do not truly reflect the wealth created by the company and are subject to manipulation by the management.

* * * END OF EXAMINATION PAPER * * *